# Routing at Network Layer

Exp no: 11 Date:

Aim: Simulate Static Routing Configuration using cisco packet Tracer.

Creating, adding, verifying static routes

→ Routers automatically learn their connected networks. We only need to add routes for the networks that are not available on the router's interfaces.

→ We also configured a backup route for network 30.0.0.0/8. The router must put the backup route to the routing table and use it to forward data packets to network 30.0.0/8 when the main route fails. To verify this, we have to simulate the failure of the main routes.

→ To simulate the failure of the main route, you can delete the link between Router0 and Router1. After deleting the link, do the same testing again for the network 30.0.0/8.

→ Network 30.0.0.0/8 and network 50.0.0/8 are not available on Router0. We have to create and add routes only for these networks.

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Router 1 requirements

* Create two routes for network 10.0.0.0/8 and configure the first route as the main route and the second route as a backup route

* Verify the router adds only 1 main route to the routing table.

Router 1 configuration:

```
Router > enable
Router# configure
terminal
Enter configuration commands, one per line
Router (config)# ip route 10.0.0.0.
Router (config)# ip route 10.0.0.0
Router (config)# exit
Router# show. ip route static
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